Individual Evidence Pack: Milestone 2

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Role: Architecture Lead & Spec Owner
Project: PolyWatch – Polymarket Integrity Monitor
Milestone: 2
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1. Key Contributions Since Milestone 1

- **Authored Threat Model v2.0:** Extended the STRIDE analysis with a new threat category **T-05 (Statistical Fabrication / Benford's Law Violation)**, motivated by empirical evidence discovered during M1 validation that fabricated bot volume exhibits non-natural leading-digit distributions. Also upgraded the impact rating of T-02 (Wash Trading) from Medium to **HIGH** after Member D's forensic case POLYWATCH-2026-001 confirmed that only 5 coordinated wallets can move a market by +15.5%.
- **Formalized Benford's Law Detection Specification:** Authored `milestone_2/specs/benford_law.feature` in Gherkin syntax to give Member C a deterministic, testable contract for the second detection algorithm. The spec covers: healthy market pass, fabricated round-number volume fail, insufficient sample size skip, configurable threshold, and boundary condition parametric tests.
- **Updated Sybil Clustering Metric (T-03):** Refined the detection rule to add block-timing correlation as a second signal (inter-wallet first-trade delta < 120 s), in addition to shared funding source. This change came from reviewing Member D's POLYWATCH-2026-001 audit log, which showed the 5 suspect wallets all made their first Polymarket trade within a 4-minute window.
- **Performed Peer Code Review of `zscore_detector.py` and `whale_alert.py`:** Ran both modules against controlled test data, verified correctness against the M1 spec, and filed inline feedback on threshold calibration (see Validation section below).
- **Formally Deprecated API-Only Data Path:** Updated the architecture constraint in Threat Model v2.0 to mandate RPC-level log parsing, closing the 40% address-masking gap discovered in M1. This aligns the data pipeline spec with Member B's current implementation direction.

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2. Verifiable Evidence

Exhibit A — Threat Model v2.0 (`milestone_2/threat_model_v2.md`)

A living security analysis document that captures the evolution of our threat understanding. The diff from v1.0 includes:

Change	Location in File
Added **T-05** (Benford's Law) with χ^2 formula	Section 3 — Threat Table

Change	Location in File
Upgraded T-02 Impact to **HIGH**	Section 3 — Threat Table
Added block-timing correlation to T-03	Section 3 — Threat Table
Added "The Fabricator" attacker profile	Section 2 — Attacker Profiles
Added Priority Order for MVP	Section 4 — Risk Matrix
Added `BenfordAnalyzer` architecture constraint	Section 5 — Architecture Constraints

> **Link:** [`milestone_2/threat_model_v2.md`]

Exhibit B — Benford's Law Gherkin Specification (`milestone_2/specs/benford_law.feature`)

A new executable specification file extending the spec framework established in M1. This file defines the contract for Member C's `BenfordAnalyzer` implementation.

Summary of scenarios defined:

Scenario	Input	Expected Outcome
Healthy market (organic data)	350 trades, natural distribution	`BENFORD_PASS`, no alert
Fabricated round-number volume	352 trades, clustered at 1K/2K/5K/10K	`BENFORD_FAIL`, HIGH alert
Insufficient sample	45 trades	`BENFORD_SKIP`, no alert
Configurable threshold	Analyst sets p=0.05 df=4	Alert at chi2 > 9.488
Boundary parametric tests	chi2 ∈ {15.506, 15.507, 15.508, 40.0}	Correct PASS/FAIL at boundary

> **Link:** [`milestone_2/specs/benford_law.feature`]

Exhibit C — Peer Review: Algorithm Validation Output

I manually executed both core analysis modules against controlled synthetic data to verify they conform to the M1 spec before they are integrated into the MVP pipeline.

Environment: Python 3.10.6, PolyWatch venv, `numpy==1.x`, `pandas==2.x`
Command equivalent: `python -m core_analysis.zscore_detector` and `python -m core_analysis.whale_alert`

Output (verbatim):

```
ZScoreDetector — Anomalies detected: 4
          price   z_score  is_anomaly
74    97.380255 -2.553061        True
106  101.886186  2.590341        True
113  102.463242  2.539027        True
120  110.791032  3.939577        True

WhaleAlert — Whale trades detected: 3
   trade_id  trade_size
2         2         300
5         5         410
8         8         500
```

Observation: The injected +10 spike at index 120 produces a z-score of **3.94** (well above the 2.5 threshold) and is correctly flagged. The three trades above the $200 whale threshold (300, 410, 500) are all correctly captured.

> **Files reviewed:** [`core_analysis/zscore_detector.py`](core_analysis/zscore_detector.py), [`core_analysis/whale_alert.py`](core_analysis/whale_alert.py)

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3. Validation Performed

Objective: Verify that `ZScoreDetector` and `WhaleAlert` conform to the behavior defined in the M1 specification (`specs/wash_trading.feature` and the Whale Alert definition in `分工.md`).

Method:

1. Seeded a 200-point synthetic price series (`np.random.seed(42)`, mean=100, std=1).
2. Injected a controlled +10σ price spike at index 120 to simulate a pump event.
3. Instantiated `ZScoreDetector(window=20, threshold=2.5)` and called `.detect()`.
4. Separately tested `WhaleAlert(size_threshold=200)` against a 10-trade sample covering sizes below and above threshold.

Result:

Test	Expected	Actual	Pass?
Injected spike at idx 120 is flagged	`is_anomaly = True`, z_score ≫ 2.5	z_score = **3.94**, flagged ✓	**PASS**
Non-spiked points not over-flagged	≤ 3 additional flags from noise	3 additional flags (idxs 74, 106, 113)	**ACCEPTABLE**
Whale threshold: trades ≥ 200 caught	3 trades (300, 410, 500)	All 3 captured ✓	**PASS**
Whale threshold: trades < 200 excluded	7 trades excluded	All 7 excluded ✓	**PASS**

Feedback filed to Member C: The 3 additional z-score flags (rows 74, 106, 113) are technically correct given `threshold=2.5` on a 200-point Gaussian series. For a production system with real market noise, a threshold of **3.0** is recommended to reduce false-positive rate. This is recorded as a spec refinement suggestion, not a bug.

4. AI Usage Transparency

Adopted:
I used an LLM to generate the initial boilerplate structure for `benford_law.feature` — specifically the `Scenario Outline` + `Examples` table for boundary condition testing. This reduced authoring time significantly. I then manually revised all numerical values (chi2 thresholds, digit counts) to be mathematically consistent with the actual Benford distribution formula $P(d) = \log_{10}(1 + 1/d)$, which the LLM had approximated incorrectly in its first draft.

Rejected:
I rejected an AI suggestion to add **"front-running via MEV bots"** as a new threat (T-05 candidate) in Threat Model v2.0. The rationale for rejection: Polymarket operates a **Central Limit Order Book (CLOB)** where trade matching is performed off-chain by a centralized sequencer, and settlement is then submitted on-chain. This architecture does not expose a public mempool at the application layer, making traditional MEV sandwich attacks non-applicable to Polymarket position-taking. The AI conflated Polymarket with Uniswap-style AMM protocols, which was technically incorrect for this system.

Appendix A — Threat Model v2.0 (`milestone_2/threat_model_v2.md`)

> Full source document embedded below for self-contained submission.

PolyWatch Threat Model (v2.0)

Project: PolyWatch - Polymarket Integrity Monitor
Owner: WU Wanpeng (Member A – Architecture Lead)
Date: Mar 3, 2026
Previous Version: v1.0 (Feb 12, 2026)
Target System: Polymarket CTF Exchange (Polygon PoS Chain)

Changelog from v1.0

Change	Description
Added T-05	New threat: Statistical Fabrication / Benford's Law Violation. Motivated by empirical validation finding during Milestone 1 that the Gamma API masks ~40% of taker addresses, forcing RPC-level analysis that also revealed non-natural digit distributions in fabricated volume data.
Updated T-03	Refined Sybil clustering metric to include block-timing correlation, not just funding source. Motivated by peer discussion with Member D during forensic case study review (POLYWATCH-2026-001).
Escalated T-02 Impact	Upgraded Wash Trading impact from Medium to High after Member D's manual audit (Case POLYWATCH-2026-001) confirmed a single entity moved price by +15.5% using only 5 coordinated wallets, exceeding original impact estimate.

Change	Description
Architecture Note	Confirmed RPC-only mandate for address resolution. API-only path formally deprecated in this version.

1. System Assets

- **Market Price Integrity:** Price must reflect genuine crowd belief, not coordinated manipulation.
- **Volume Authenticity:** Reported trade volume must consist of economically meaningful, arms-length transactions.
- **User Anonymity:** Behavioral patterns of on-chain actors are sensitive; PolyWatch reads but does not expose raw PII.
- **Liquidity Provider Fairness:** LPs must not be systematically exploited by toxic, information-asymmetric order flow.

2. Attacker Profiles (Updated)

Profile	Motivation	Capital Requirement
The Whale	Move odds to influence public perception or hedge a large off-platform position.	High (>$500K)
The Farmer	Farm volume-based rewards / airdrops through wash trading to generate fee rebates.	Low–Medium
The Insider	Trade on material non-public information before market resolution.	Medium
The Fabricator *(New v2.0)*	Manufacture synthetic order history to make a thin market appear liquid, attracting real counterparties.	Low (off-chain data manipulation risk)

3. Threat Analysis (STRIDE Methodology) — Updated

ID	Threat Category	Attack Vector	Technical Description	Impact	Detection Metric
T-01	**Spoofing**	**Layering / Phantom Orders**	Attacker places deep OTM limit orders to create a false "Buy Wall", then cancels them before execution.	Misleads retail traders; induces FOMO buying.	**Cancellation Ratio:** `(Cancelled_Volume / Executed_Volume) > 10.0` within 5 mins.

ID	Threat Category	Attack Vector	Technical Description	Impact	Detection Metric
T-02	**Tampering**	**Wash Trading (Circular)** *(Impact upgraded)*	Accounts A→B→C (same entity) trade the same contracts back and forth to artificially inflate volume. Confirmed via POLYWATCH-2026-001: 5 wallets pumped price +15.5% with ~$2.1M in circular buys.	**HIGH** – Directly distorts settlement odds; platform reward fraud.	**Graph Cycles:** Closed loops `A → B → C → A` with net-zero inventory change. ZScore detector triggers on price delta > 2.5σ.
T-03	**Repudiation**	**Sybil Attack** *(Metric updated)*	100+ wallets funded by same Mixer simulate "grassroots" consensus. Detection now also includes: wallets with identical funding block ±3 blocks and synchronized first-trade timing < 120s.	Distorts Wisdom-of-the-Crowd; creates fake consensus.	**Funding Clustering** (Mixer source) **+ Block-Timing Correlation** (inter-wallet tx delta < 120s).
T-04	**Info Disclosure**	**Copy-Trading Leakage**	Observers monitor famous addresses to front-run positions.	Privacy loss for high-profile traders.	**Time Correlation:** Target_Wallet tx → Follower_Wallet tx within 30s.

ID	Threat Category	Attack Vector	Technical Description	Impact	Detection Metric
T-05	**Tampering**	**Statistical Fabrication** *(New v2.0)*	Fabricated or bot-generated trade volumes exhibit non-natural leading-digit distributions, violating Benford's Law. Genuine organic activity follows $P(d) = \log_{10}(1 + 1/d)$. Significant deviation ($\chi^2 > 15.507$) indicates synthetic volume.	Attracts real liquidity to an artificially deep market; enables exit scam by fabricator.	**Benford's Law Test:** $\chi^2 = \Sigma (O\_d - E\_d)^2 / E\_d$, where $E\_d = N \cdot \log_{10}(1 + 1/d)$. Flag if $\chi^2 > 15.507$ (df=8, p=0.05).

4. Risk Matrix (Updated)

```
          |  LOW Impact   |  MED Impact   |  HIGH Impact
──────────┼───────────────┼───────────────┼──────────────
HIGH      |               |  T-03, T-05   |  T-02
Likelihood────────────────────────────────────────────────
MED       |  T-04         |  T-01         |
──────────┼───────────────┼───────────────┼──────────────
LOW       |               |               |
```

Priority order for MVP detection engine: T-02 → T-03 → T-05 → T-01 → T-04

5. Architecture Constraints Derived from This Threat Model

1. **RPC-Mandatory:** Gamma API alone is insufficient (40% address masking confirmed in M1). All address-level analysis requires raw `CTFExchange` log parsing via Polygon RPC.
2. **Graph Database Needed for T-02/T-03:** NetworkX (in-memory) is acceptable for MVP; migrate to Neo4j for production-scale Sybil clustering.
3. **Statistical Module Needed for T-05:** Implement `BenfordAnalyzer` class in `core_analysis/` with configurable `chi2` threshold; threshold defaults to `15.507` (p=0.05, df=8).

Appendix B — Benford's Law Specification (`milestone_2/specs/benford_law.feature`)

> Full source document embedded below for self-contained submission.

```gherkin
# language: en
Feature: Benford's Law Anomaly Detection for Trade Volume Fabrication
  As a Market Integrity Analyst
  I want to test whether trade volume digit distributions follow Benford's Law
  So that I can identify markets with statistically fabricated transaction
volumes

  # This specification drives the implementation of BenfordAnalyzer in
core_analysis/ (Member C)
  # Mathematical basis: P(d) = log10(1 + 1/d) for leading digit d in {1..9}
  # Significance test: chi-square goodness-of-fit, df=8, critical value=15.507 at
p=0.05

  Background:
    Given the "BenfordAnalyzer" module is initialised with chi2_threshold =
15.507
    And a minimum sample size of 100 trades is required for the test to be valid

  Rule: Organic trade data must conform to Benford's Law digit distribution

  Scenario: Healthy market passes the Benford test
    Given a market "BTC Price > 100k by Dec 2024" with 350 historic trade records
    When the leading digits of the "trade_size_usd" field are extracted
    And the chi-square goodness-of-fit test is run against the Benford expected
distribution
    Then the resulting chi2 statistic should be less than 15.507
    And the market should be labelled "BENFORD_PASS"
    And no alert should be raised

  Scenario: Fabricated round-number volume fails the Benford test
    Given a market "Lakers Win 2026 NBA Championship" with 352 trade records
    And the trade sizes are clustered around round numbers [1000, 2000, 5000,
10000]
    When the leading digit distribution is computed:
      | Leading Digit | Observed Count | Benford Expected % |
      | 1             | 12             | 30.1%              |
      | 2             | 95             | 17.6%              |
      | 3             | 8              | 12.5%              |
      | 4             | 7              | 9.7%               |
      | 5             | 93             | 7.9%               |
      | 6             | 9              | 6.7%               |
      | 7             | 8              | 5.8%               |
      | 8             | 9              | 5.1%               |
      | 9             | 111            | 4.6%               |
    And the chi-square statistic is computed as sum((O - E)^2 / E) over all
digits
    Then the chi2 statistic should be greater than 15.507
    And the market should be flagged as "BENFORD_FAIL"
    And the system should emit an alert of level "HIGH" with reason "Potential
Volume Fabrication"
    And the specific market_id and chi2_value should be logged to the
"Suspicious_Activity" table

  Scenario: Insufficient sample size – test is skipped
    Given a market "Will Elon Musk win the 2030 election?" with only 45 trade
records
```

```gherkin
    When the BenfordAnalyzer attempts to run
    Then the test should be skipped
    And the market should be labelled "BENFORD_SKIP – Insufficient Data (n=45, min=100)"
    And no alert should be raised

  Rule: The chi-square threshold is configurable for sensitivity tuning

  Scenario: Analyst tightens the threshold for high-stakes markets
    Given the "BenfordAnalyzer" is reconfigured with chi2_threshold = 9.488 (p=0.05, df=4 – first 5 digits only)
    When a market with chi2 = 11.2 is evaluated
    Then the market should be flagged as "BENFORD_FAIL"
    And the alert metadata should record which threshold value was used

  Scenario Outline: Parametric threshold boundary conditions
    Given a market with chi2 statistic of <chi2_value>
    And the configured threshold is 15.507
    Then the detection result should be <expected_result>
    Examples:
      | chi2_value | expected_result |
      | 15.506     | BENFORD_PASS    |
      | 15.507     | BENFORD_PASS    |
      | 15.508     | BENFORD_FAIL    |
      | 40.0       | BENFORD_FAIL    |
```